Test and Evaluation (Part I)

3.5.5

SEAPOWER THROUGH ENGINEERING

TOPIC LEARNING OBJECTIVES

Upon successful completion of this topic, the student will be able to:

1. Recognize the fundamentals of Test and Evaluation (T&E).
2. Identify how T&E supports risk management by milestone decision authorities.
3. Recognize how T&E supports the Systems Engineering Process.
4. Identify the characteristics and purposes of the different types of T&E and the organizations responsible for each.
5. Identify the key T&E support organizations within DoD.
6. Identify the role of modeling and simulation as a tool in the systems engineering process.
7. Recognize the importance of modeling and simulation in the defense acquisition process.
8. Identify the regulation that dictates the development of a Test and Evaluation Master Plan (TEMP) and recognize why there is a need to develop a TEMP for most programs.
9. Recognize how the TEMP generation, staffing and approval process integrates all functional disciplines throughout the acquisition life-cycle.
10. Identify issues affecting T&E resource requirements, test planning, and test execution activities in support of a program's acquisition strategy.
11. Given descriptions, identify the types of Development Test and Evaluation (DT&E), e.g., Production Qualification Tests and Production Acceptance Test and Evaluation.

STUDENT PREPARATION

Student Support Material

1. None

Primary References

1. DoD 5000 Series
2. DoDI 5000.89 Test and Evaluation
3. SECNAVINST 5000.2G
4. T&E Enterprise Guidebook

Additional References

1. DoDM 5000.100 Test And Evaluation Master Plans And Test And Evaluation Strategies

TOPIC LEARNING OBJECTIVES

Upon successful completion of this topic, the student will be able to:

12. Recognize how Technical Performance Measures are used to track progress in program risk areas during systems development.
13. Identify which organizations develop, coordinate, and approve Critical Technical Parameters (CTPs).
14. Identify key issues regarding test and evaluation of commercial and non-developmental items (NDI).

STUDENT PREPARATION

Student Support Material

1. None

Primary References

1. DoD 5000 Series
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3. SECNAVINST 5000.2G
4. T&E Enterprise Guidebook

Additional References

1. DoDM 5000.100 Test And Evaluation Master Plans And Test And Evaluation Strategies

Overview

- Introduction
- T&E Organization
- Developmental T&E Types
- T&E Terminology
- Test and Evaluation Master Plan (TEMP)
- Modeling and Simulation

What is T&E?

- Test and Evaluation (T&E) is the process by which a system or components are compared against requirements and specifications through testing. The results are evaluated to assess progress through design, performance, and supportability
- T&E exercises a system or component(s) and analyzes results to provide performance-related information to:
    - Identify risks
    - Reduce cost, schedule, and performance risk
    - Validate models & simulation
    - Determine effectiveness, suitability, survivability, and/or lethality
    - Demonstrate exit criteria needed for milestones and decisions
    - Help predict program deviations
    - Assist in trade studies

The fundamental purpose of T&E is to enable the DoW to acquire systems that work

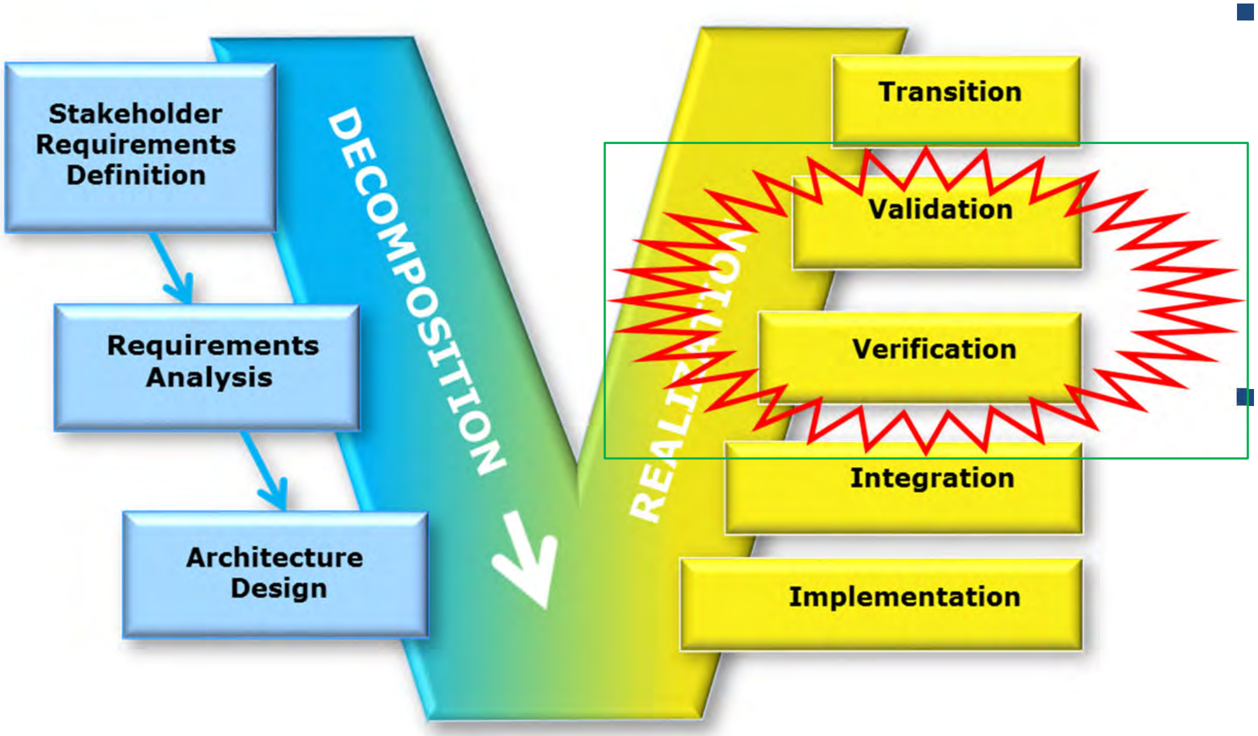
Why Test?

T&E Terminology

- Test – a program, procedure, or process to obtain, verify, or provide data for determining the degree to which a system (or subsystem) meets, exceeds, or fails to meet its stated capabilities
- Evaluation – the review, analysis, and assessment of data obtained from testing used to project system performance under operational conditions
- Difference between Testing and Evaluation:

Testing	Evaluation
Obtains raw data	Produces analyzed information from test data, modeling and simulation, or other sources
Measures specific, individual performance factors	Draws conclusions by looking at how the data interact
Is resource intensive	Is intellectually intensive

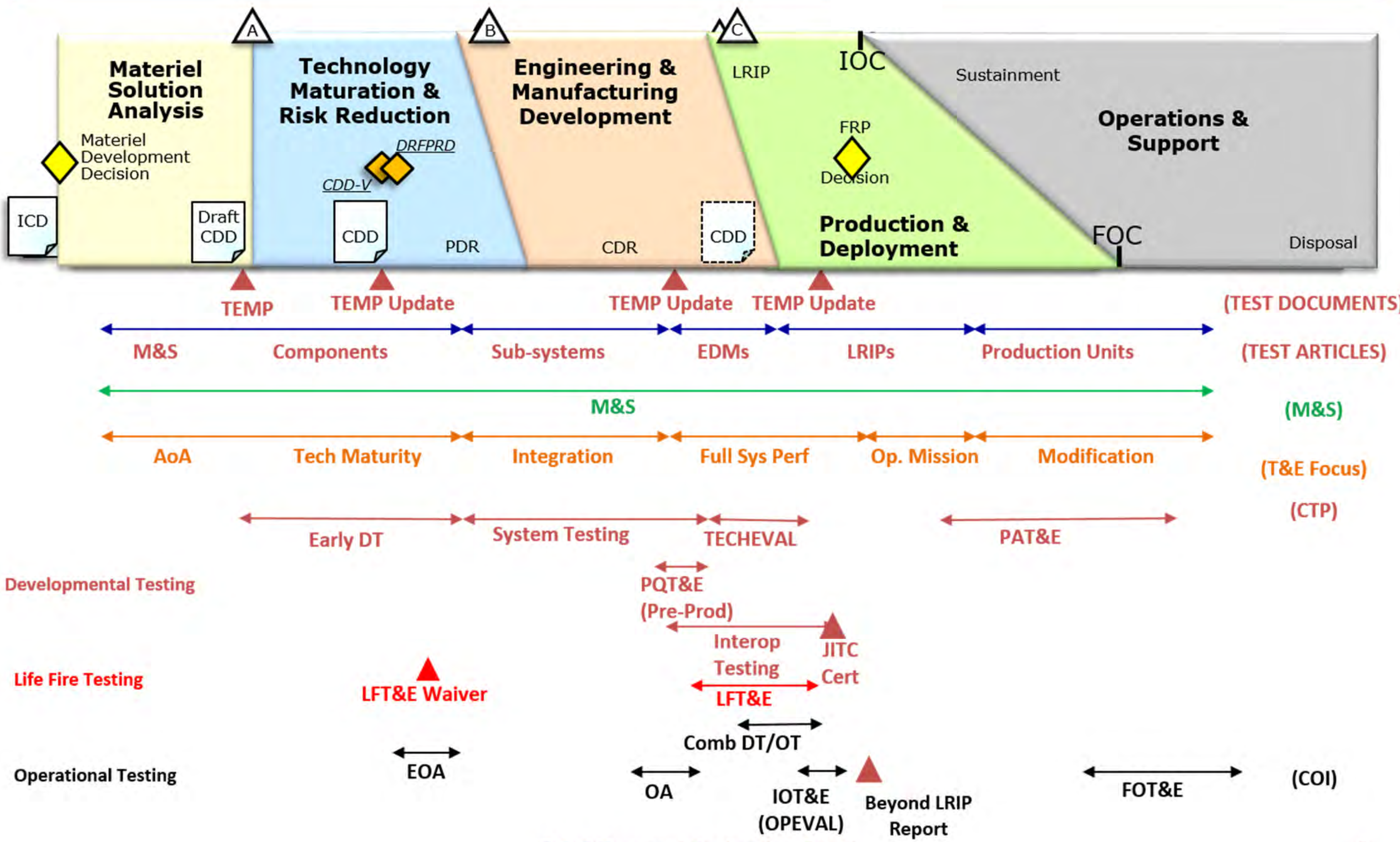
Systems Engineering Process



- ***Verification*** – ensures that the solution meets the specified requirements
  - Did we build it right?
- ***Validation*** – ensures that the solution meets the user's needs
  - Did we build the right thing?

T&E is the primary method to verify the design meets requirements and validate solution meets user's needs

Test and Evaluation



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Statutory Requirements

- WEAPON SYSTEMS ACQUISITION REFORM ACT of 2009 (WSARA 2009):
  - SEC. 102
    - Created position of Director of Developmental Test and Evaluation (now Deputy Assistant Secretary of War for Developmental Test & Evaluation (DASW DT&E))
  - SEC. 104
    - Requires periodic reporting by DASW DT&E on technological maturity and integration risk of critical technologies of the major defense acquisition programs
  - Director of Defense Research and Engineering (DDR&E) submits an annual report to Congressional Defense committees describing additional resources that may be required for:
    - The annual report on the technological maturity and integration risk of critical technologies of major defense acquisition programs
    - The technological maturity assessments required by section 2366b(a) (renumbered 4251 and 4252) of Title 10, United States Code
    - The requirements of DoDI 5000, as revised

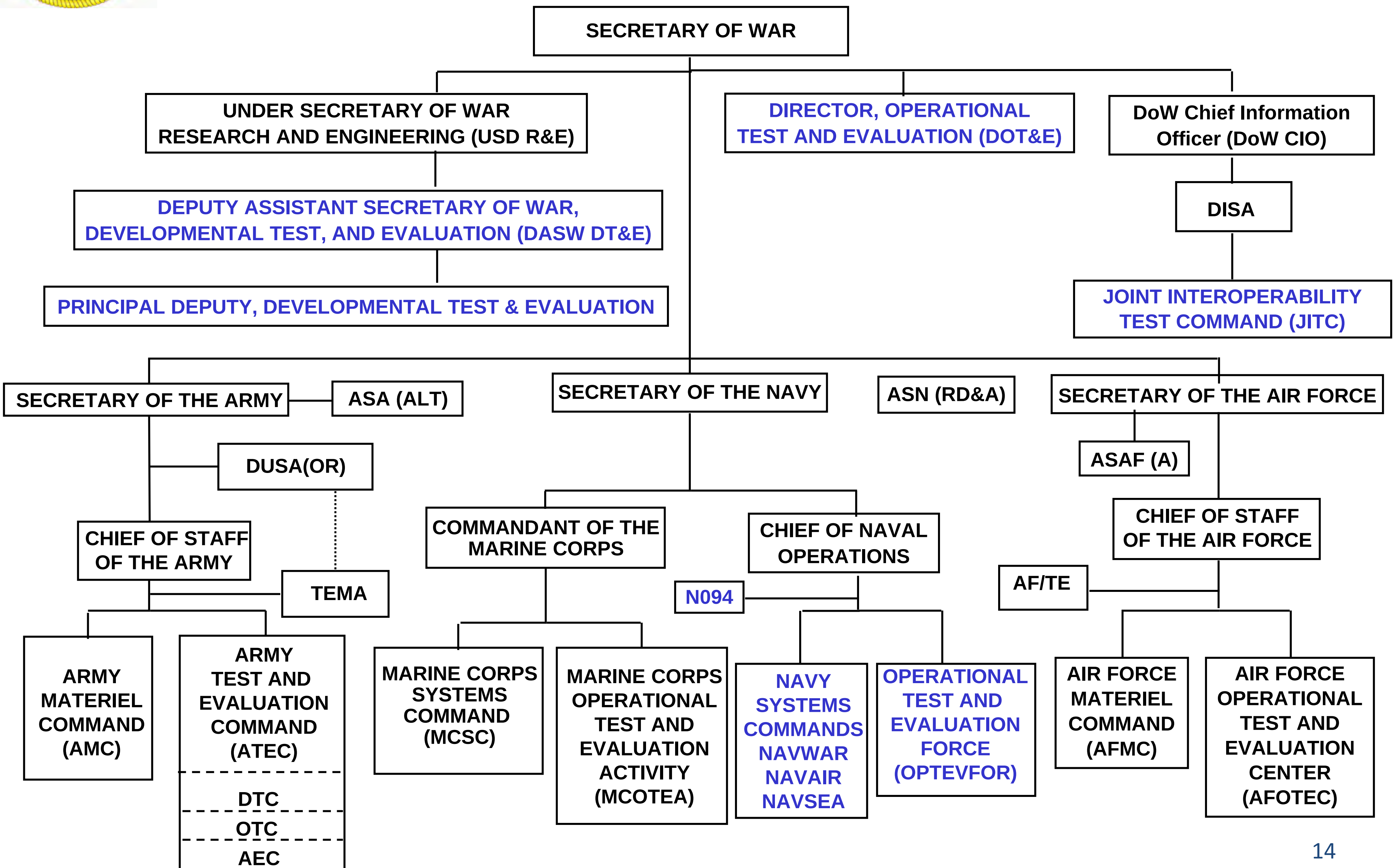
Statutory Requirements

- WEAPON SYSTEMS ACQUISITION REFORM ACT of 2009 (WSARA 2009):
  - DDR&E, in consultation with the DASW DT&E:
    - Shall develop knowledge-based standards against which to measure the technological maturity and integration risk of critical technologies at key stages in the acquisition process for purposes of conducting the reviews and assessments of major defense acquisition programs required by subsection (c) of section 139a of Title 10, United States Code
  - Secretary of War approves Independent Technical Risk Assessments (ITRA) before milestone A or B per sections 4251 and 4252, respectively, of title 10, U.S. Code. 10 U.S.C. §4272
    - Secretary of War delegated this responsibility, to OUSW(R&E) per DoDI 5000.88, Engineering of Defense Systems, 18 Nov 2020
- 10 U.S.C. 2366a (renumbered 4251) Major defense acquisition programs; certification required before Milestone A approval
- 10 U.S.C. 2366b (renumbered 4252) Major defense acquisition programs; certification required before Milestone B approval
- 10 U.S.C. 2399 (renumbered 4171) Operational test and evaluation of defense acquisition programs

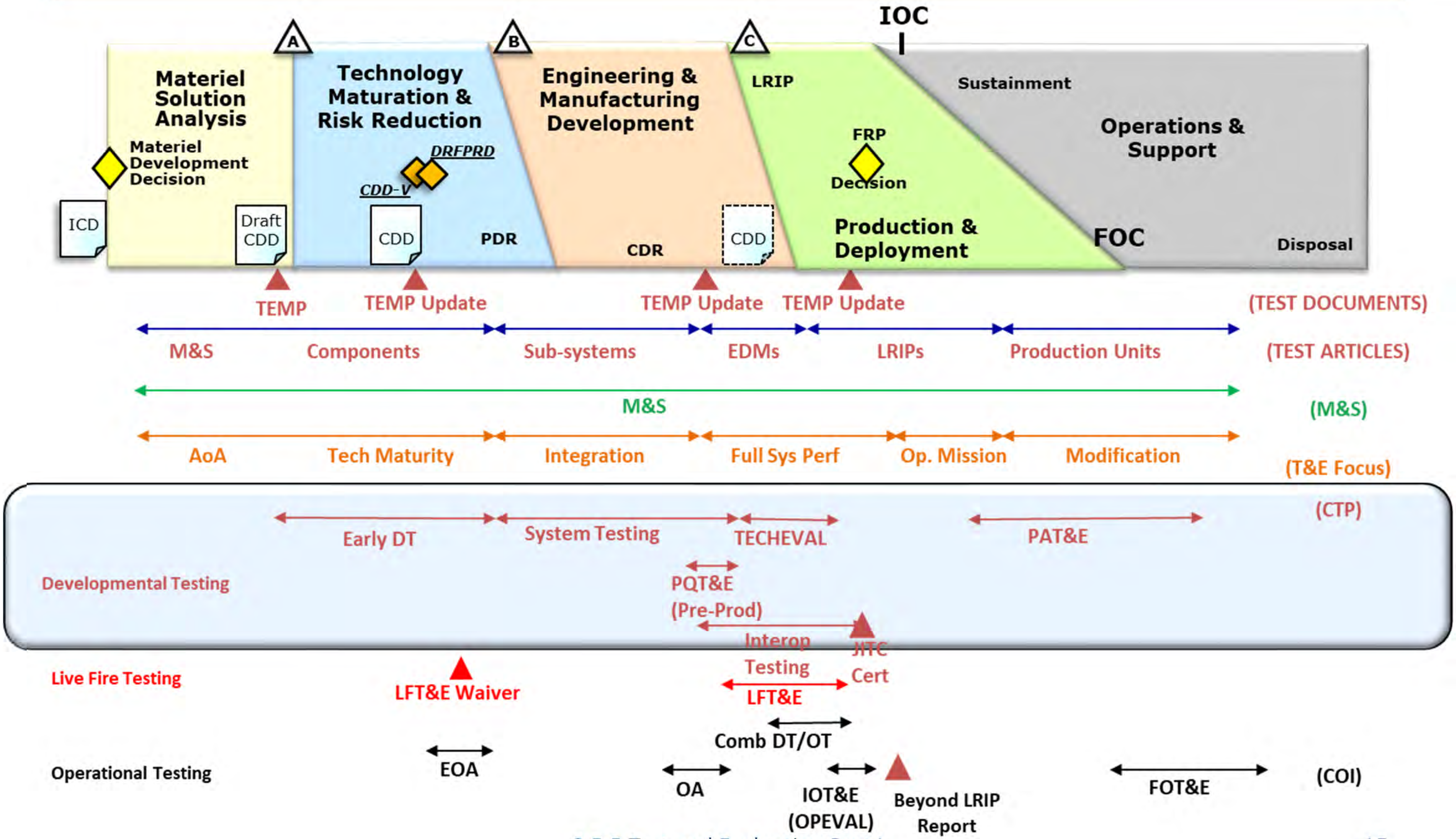
Deputy Assistant Secretary of War for Developmental Test & Evaluation (DASW DT&E)

- Responsibilities outlined in Title 10, United States Code
- Principal advisor to USW (A&S) for DT&E-related matters
  - Responsible for evaluating DT&E capability within DoW
    - Program oversight
    - Policy & guidance
    - Test & Evaluation Master Plan (TEMP) approval (with DOT&E)
  - Collaborative partner with DOT&E to ensure acquisition decisions are supported with appropriate T&E information
  - Partner with Component Acquisition Executive (CAE) to ensure each acquisition program is supported by an appropriate DT&E Strategy
  - Advocate for DT&E Acquisition workforce
  - Prepares annual report to Congress

DoW T&E Organization



Test & Evaluation



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Developmental Test and Evaluation (DT&E)

- Conducted throughout the acquisition process to assist in engineering design and development
- Used to verify component and system technical performance specifications have been met
- Controlled by Government PMO
- Performed (typically) by Government labs (e.g., NSWC) and/or Contractor teams
- Helps identify & control design technical risk
- DASW DT&E may oversee
- DT&E Types:
  1. System Testing (verifying performance requirements)
  2. PQT&E: Production Qualification T&E
     a. Pre-production (prior to M/S C)
     b. 1st Article (between M/S C and FRP)
  3. PAT&E: Production Acceptance T&E

Ensures the producibility of the system's design and the effectiveness of the manufacturing process

1. System Testing & Evaluation

- Testing performed during development that verifies performance
  – Typical development process is design-build-test and repeat
- Identifies and controls design technical risk
  – Assess progress toward resolving critical technical and operational issues, mitigation of tech risk, achievement of manufacturing process requirements, and system maturity
- Provides feedback to SE Process
  – Verifies requirements and validates functions
  – Supports cost, schedule and performance trade-offs with data
- Provides data and analysis to confirm the system is ready for OT&E
  – Want reasonable assurance of success before committing resources for OT

2. Production Qualification T&E (PQT&E)

- Pre-Production Qualification tests
  - Conducted prior to M/S C
  - Confirms system design integrity over specified operational & environmental range (e.g., thermal & vibration)
  - Usually on prototype or pre-production hardware (e.g., Engineering Development Model (EDM))
  - Includes contractual reliability & maintainability demonstration tests
- 1st Article Tests
  - Conducted between M/S C and Full Rate Production. (Ships are different)
  - Ensures effectiveness of manufacturing processes, equipment, and procedures on items from the first production lot
  - PMO conducts additional 1st Article Tests when manufacturing process or system design changes significantly or a second source starts production

PQT&E verifies the design and the manufacturing processes prior to FRP

DT&E Example

Pre-production Qualification Testing and Evaluation of the Enhanced Position Location Reporting System (EPLRS)

3. Production Acceptance T&E (PAT&E)

- Conducted during production, fielding/deployment, and operational support
  - Responsibility of the developer (Contractor)
  - May be accomplished by Quality Assurance sampling
- Checks that production units meet requirements/specifications

Reduces production risk

Commercial & Non-Development Items (NDI)

- Adequate testing is still required for Commercial Off-The-Shelf (COTS) & NDI
- Key issues:
  - Undefined COTS/NDI architecture
    - May impose additional risk due to lack of technical data
  - System suitability
    - Drive to maximize advantages of COTS/NDI may ignore legal requirements and suitability concerns
- Considerations:
  - Intended environment of the NDI
  - Additional integration testing
  - Potential modifications required to conform to new system
  - Take advantage of previous testing
  - Involve the Services’ DT&E and OT&E agencies
  - Thoroughly document COTS & NDI testing in TEMP

Technical Evaluation (TECHEVAL)

- Navy conducts additional DT&E for IOT&E cert called technical evaluation (TECHEVAL)

- TECHEVAL**:** The study, investigations, or Test and Evaluation (T&E) by a developing agency to determine the technical suitability of materiel, equipment, or a system, for use in the Military Services
  - DT&E controlled by the program office in a more operationally realistic test environment

The schedule should allow sufficient time between DT&E and IOT&E for rework, reports, analysis, and developmental testing of critical design changes

Technical Evaluation (TECHEVAL)

- Objectives:
  - To assist the developers by providing information relative to technical performance, qualification of components, compatibility, interoperability, vulnerability, lethality, transportability, reliability, affordability, maintainability, manpower and personnel, system safety, integrated logistics support, correction of deficiencies, accuracy of environmental documentation, and refinement of requirements
  - To ensure the effectiveness of the manufacturing process of equipment and procedures through production qualification testing
  - To confirm ***readiness for IOT&E***

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T&E Terminology

- Generic terminology for "metric"
  - TPM - Technical Performance Measure
- Developmental testing terminology
  - CTP - Critical Technical Parameter
- Operational testing terminology
  - COI – Critical Operational Issue
  - MOE – Measure of Effectiveness
  - MOS – Measure of Suitability
  - CBTE – Capabilities Based Test & Evaluation
  - MBTE – Mission Based Test & Evaluation
  - PMT – Platform Mission Task
  - IEF – Integrated Evaluation Framework

Technical Performance Measure (TPM)

- PMs must manage 3 basic program elements: cost, schedule, and performance
  - TPMs provide the mechanism for management of technical performance risk
- TPM is an umbrella term that covers Critical Technical Parameters (CTPs) and Critical Operational Issues (COIs)
  - CTPs and COIs are derived from KPPs
- Technical Performance must be tracked throughout system development
  - Results are compared to predicted values
  - Provides early detection of performance problems
  - Includes risk thresholds – when do we "pull the trigger"? (e.g., switch to alternative design or technology)

TPMs are essential to risk management

Critical Technical Parameters (CTPs)

- CTPs are ***design parameters*** critical to ensuring system ***meets operational thresholds*** (i.e., speed, range, lethality)
  - CTPs listed in matrix format in Part I of TEMP
- CTPs are TPMs derived from:
  - Capability Documents (ICD and CDD)
  - System Specifications
  - Acquisition Program Baseline (APB)
  - Systems Engineering documents
- CTPs are listed in a matrix, along with performance objectives and thresholds, in Part III of the TEMP
- CTPs are stated as specific numerical values and may change as the system matures during development; the system should show improvement over time

PMO develops and coordinates CTPs; MDA approves

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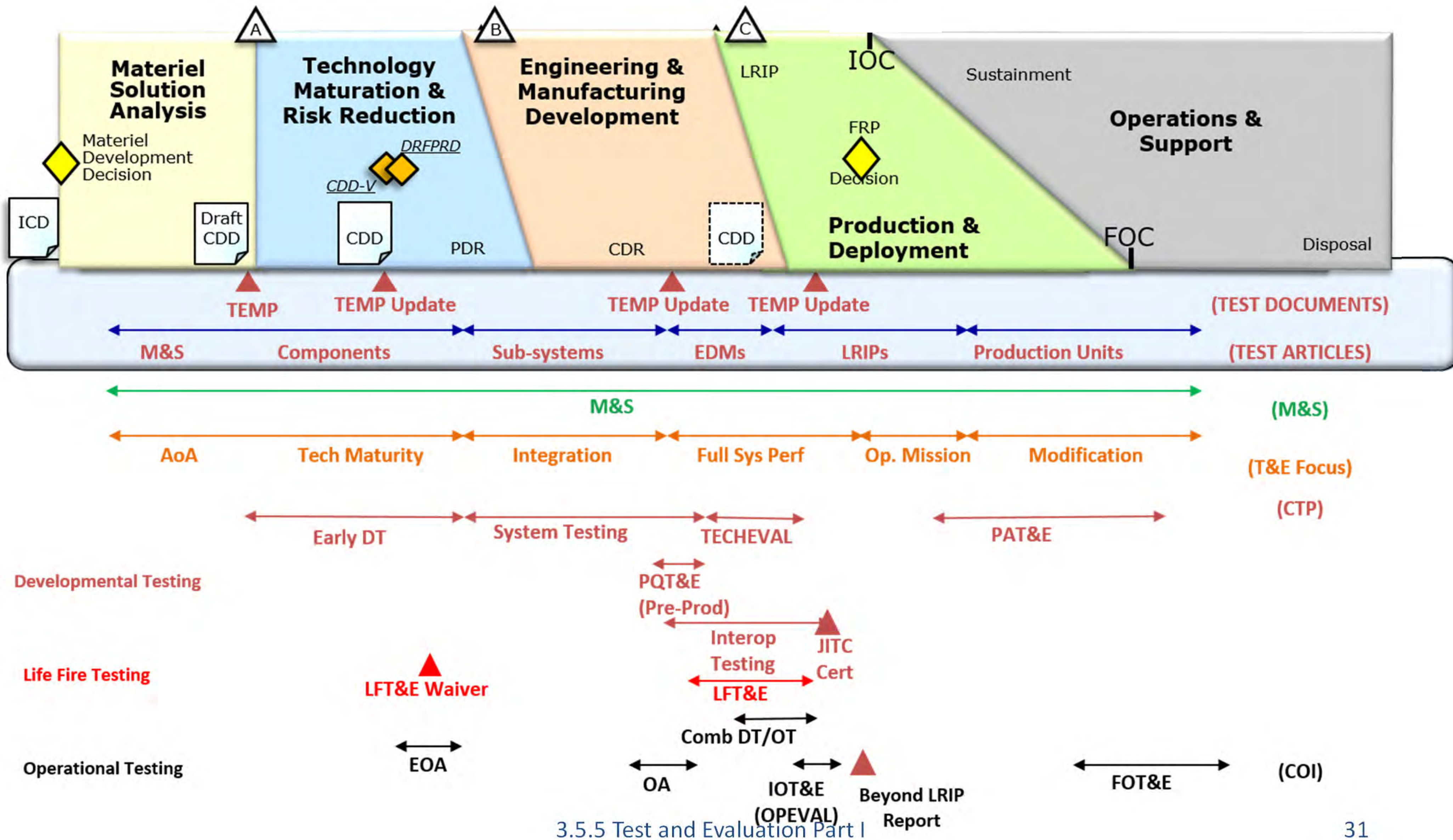
Test & Evaluation



Test & Evaluation Master Plan (TEMP)

- Provides an overall test management plan and framework within which detailed T&E plans are contained
- Must:
  - Integrate T&E with the overall Acquisition Strategy, reflect the user's capability needs and describe how these capability needs will be tested in DT&E, LFT&E, and OT&E
  - Document the T&E program for the entire life-cycle
  - Specify personnel, funding, and test range support requirements
  - Be developed prior to M/S A Review and updated before each subsequent Milestone Decision Review
  - Must include strategy and resources for cybersecurity T&E
- May be waived or substituted by another tailored test strategy document for certain acquisition pathways
  - In cases where the TEMP is not needed, early briefings to Service stakeholders, as well as USD (R&E)
- DoDI 5000.89 is the guiding document for TEMP development

TEMP Source Documents

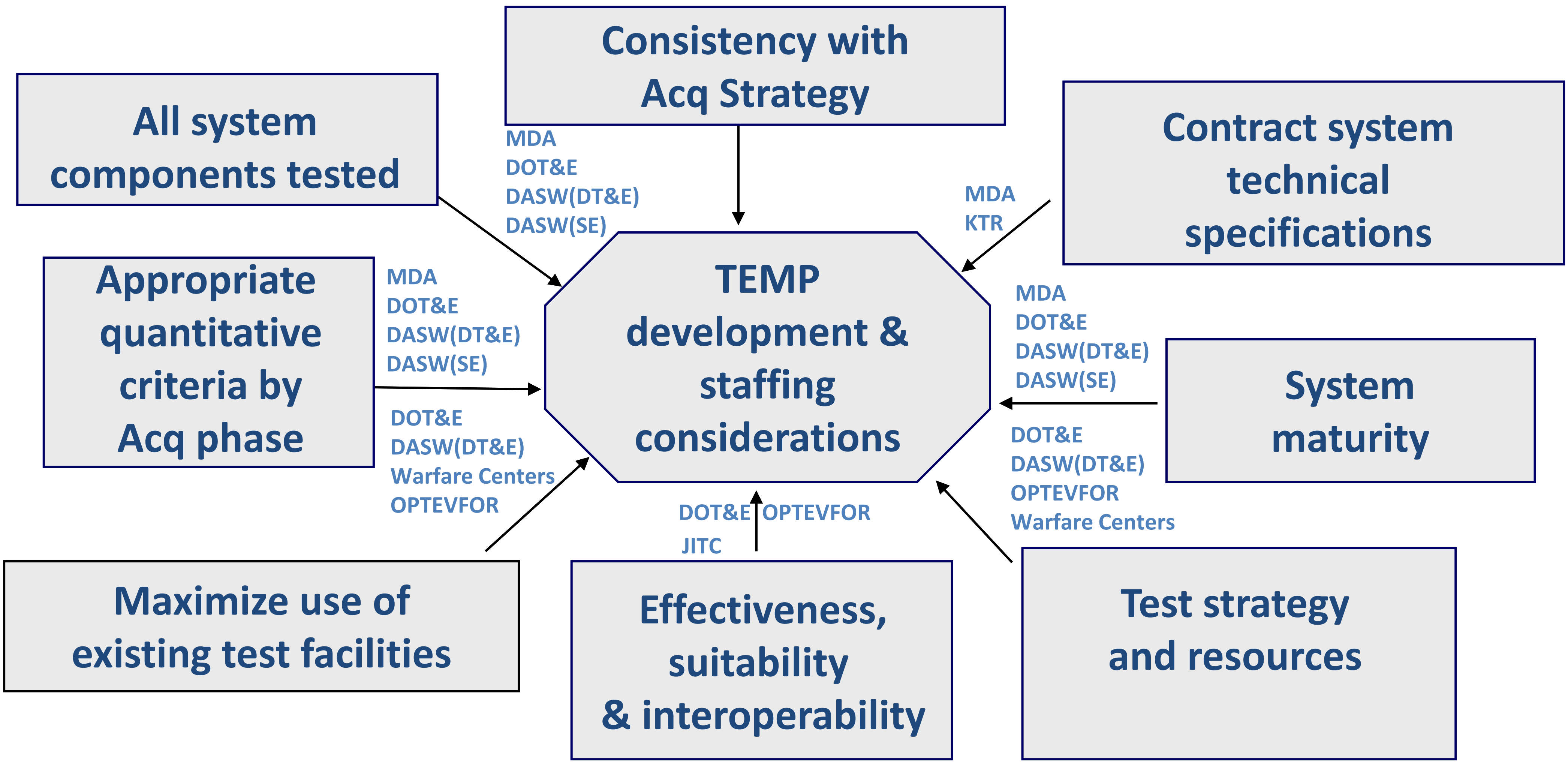
- AcqStrat - Acquisition Strategy
- AoA - Analysis of Alternatives
- APB - Acquisition Program Baseline
- ICD - Initial Capabilities Document
- CDD - Capability Design Document
- VOLT - Validated Online Life-cycle Threat assessment

<u>Capability Documents</u> *are the prime sources of testable parameters for OT&E*

Mandatory TEMP Format

- Part I - System Introduction
- Part II - Test Program Management and Schedule
- Part III - Test and Evaluation Strategy
- Part IV - Test and Evaluation Resource Summary

TEMP Planning Considerations



Numerous considerations must be balanced for T&E planning and execution. This is best accomplished via an IPT approach throughout the acquisition life-cycle

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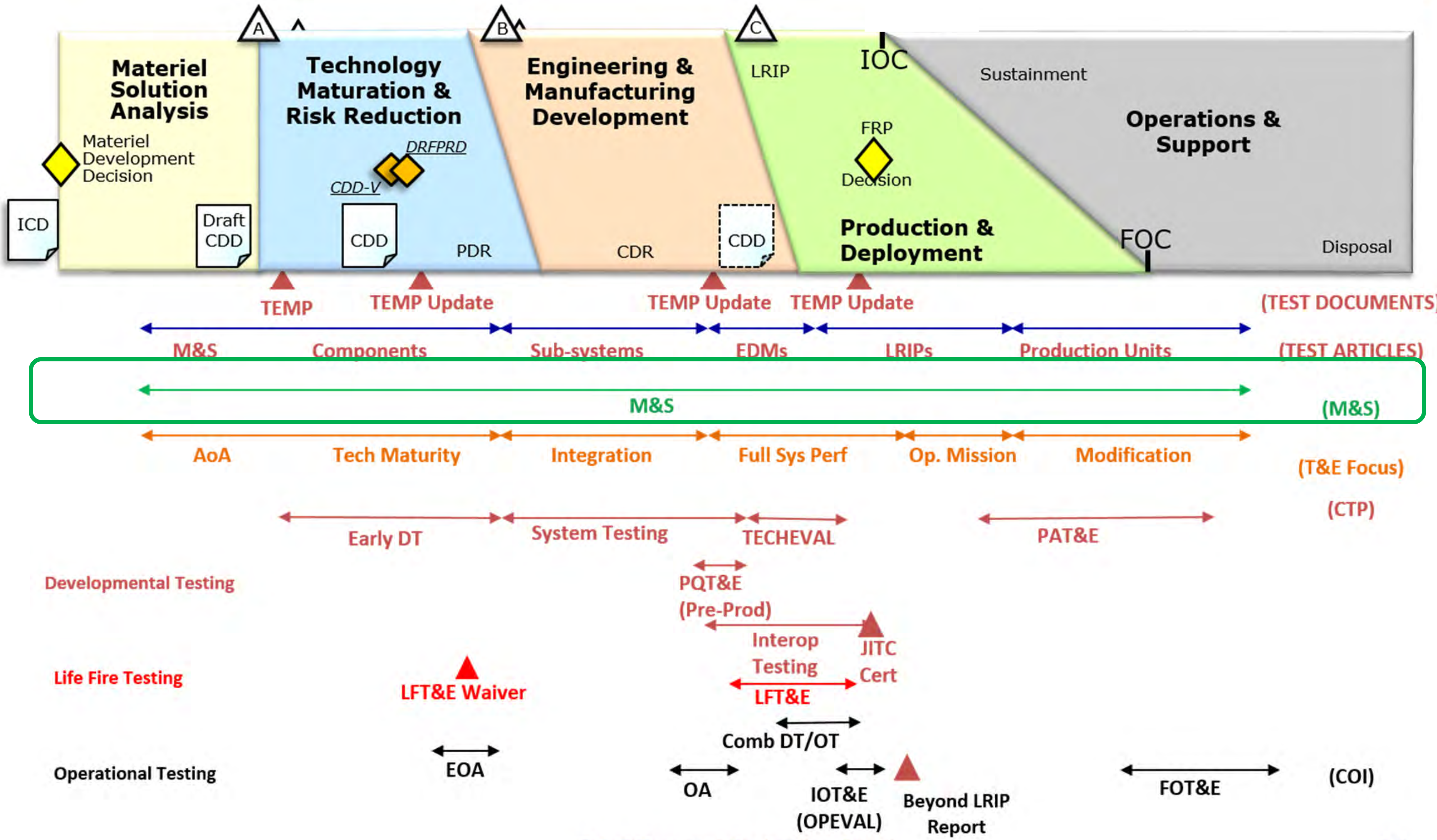


What is Modeling and Simulation?

- Model – ***<u>A representation of an actual or conceptual system</u>*** that can be used to predict how the system might perform or survive under various conditions
- Simulation – ***<u>A method for implementing a model</u>***. Conducting experiments with a model for the purpose of understanding the behavior of the system under selected conditions or of evaluating various strategies for the operation of the system. Military exercises and war games are also simulations
- Some M&S Types
  - Software in the loop (SWIL)
  - Hardware in the loop (HWIL)
  - Human-in-the-loop (HITL)
  - Constructive simulation

Model - static representation of system or process
Simulation - dynamic application of model(s) through time

Modeling and Simulation Example

M&S in Systems Engineering

- M&S can represent the system-of-systems environment for systems engineering to design, develop, and test individual systems
  - Cost precludes the development of full-scale prototypes merely to provide proof of concept
  - Cost of testing events limits the number of tests that can be conducted
  - Do not rely solely on M&S, actual testing must be conducted to catch potential problems
- M&S Through the Acquisition Phases
  - **Pre-Acquisition Concepts, Experimentation, and Prototyping** - Use M&S to: Model CONOPS, perform cost, schedule, and performance trades, discover system interoperability
  - **Materiel Solution Analysis Phase** – Use M&S to: Assess materiel solutions, estimate life cycle costs, model CONOPS and mission context, model interoperability and warfighter integration, conduct industrial and manufacturing capability analysis, model supportability and sustainment
  - **Technology Maturation and Risk Reduction Phase** - Use M&S to: Conduct trade studies, model system threat integration, model environment and demonstrate technology, conduct interoperability and supportability analysis, model operational suitability and affordability, assess industrial and manufacturing readiness, estimate manpower and cost, model system-to-performance specifications, plan T&E activities, prototype human interfaces

(source DON MSSP Guidebook)

M&S in Systems Engineering

- M&S through the Acquisition Phases (cont.)
  - **Engineering, Manufacturing, and Development Phase** - Use M&S to: Model operational supportability, model logistics footprints, analyze survivability, model human systems integration, design for producibility, demonstrate system safety, verify functionality and performance to specifications/needs, estimate manpower
  - **Production and Deployment Phase** – Use M&S to: Assess industrial/manufacturing readiness, model environment, safety, and occupational health, determine military equipment valuation, model corrosion prevention and control, refine the life-cycle, sustainment plan, test production quality, V&V production configuration, conduct economic analysis
  - **Operations and Support Phase** - Use M&S to: Model supply chain management, monitory performance and adjust product support, assess supportability, validate failures and determine root causes, determine system risk and hazard severity, analyze ECP impact

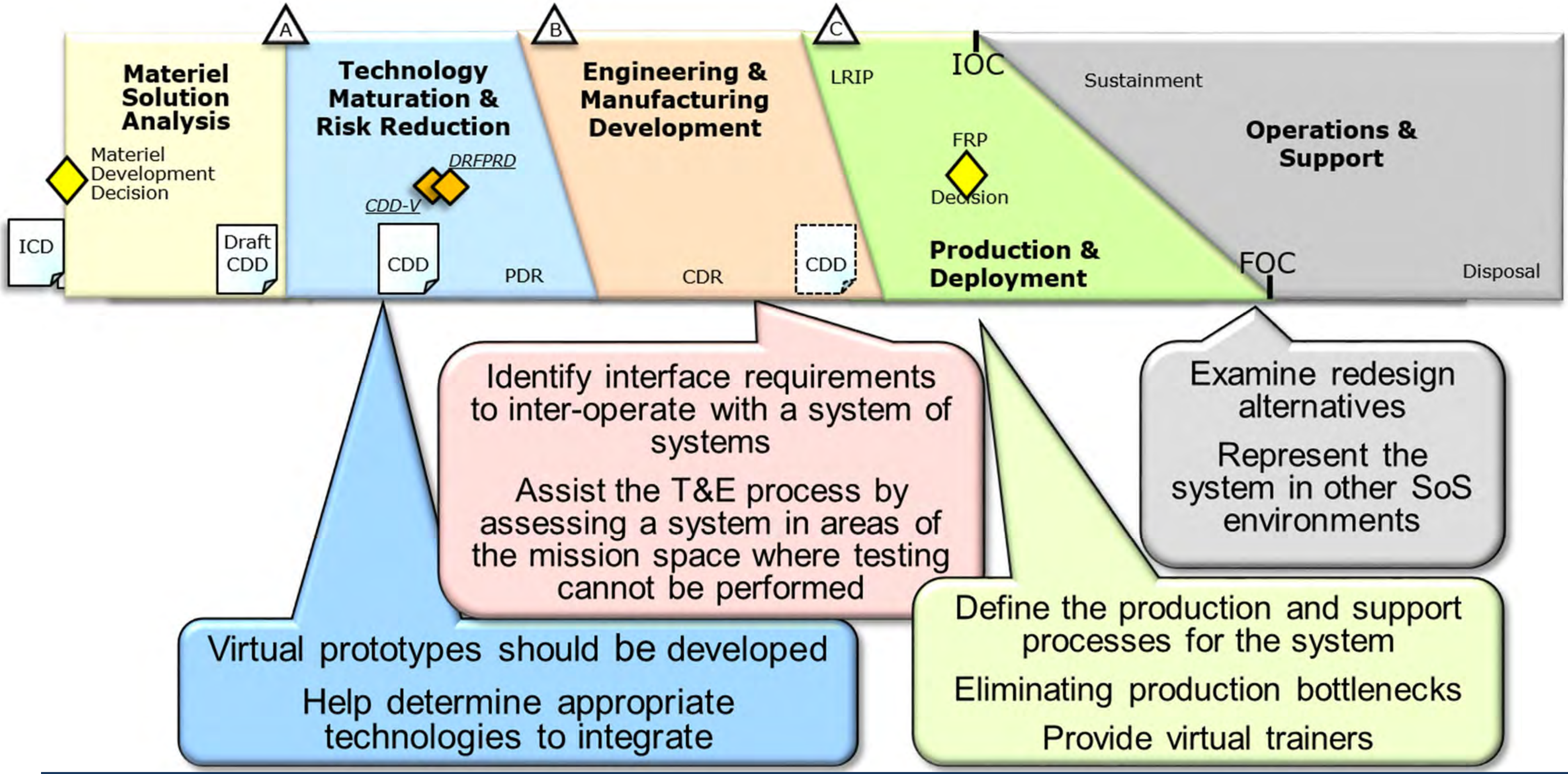
M&S supports the systems engineering decision process by supporting systems design, trade studies, financial analysis, sustainment, and performance assessments

(source DON MSSP Guidebook)

DoW Modeling and Simulation Verification, Validation, and Accreditation (VV&A)

- Verification – the model accurately represents the developer's conceptual description
- Validation – process of determining the degree to which a model is an accurate representation of the real world
- Accreditation – the official certification of M&S for a specific purpose
- DoW policy that models, simulations, and associated data used to support DoW processes, products, and decisions shall undergo verification and validation (V&V) throughout their life-cycles
- Each DoW Component shall be final validation authority for representations in common and general-use models, simulations, distributed simulations, and associated data, and shall be responsive to other DoW Components to ensure those forces and capabilities are appropriately represented
  - OPTEVFOR for Navy

M&S in Acquisition



M&S can assist acquisition by controlling variables to provide a repeatable audit trail to support decision processes

Summary

- T&E is exercising a system or components and analyzing results to provide:
- T&E supports MDA risk management by providing:
- T&E is the primary method to design meets requirements and solution meets user's needs
- Three types of T&E and responsible organizations are:
- The key T&E support organizations within the Navy are
- Key OT&E activities coordinated with DOT&E are

Summary

- Independent agencies conduct OT&E to provide:

- The Role of M&S is to:

- M&S is used throughout to support acquisition decisions